

How much power does your local llm rig use?

Posted by u/No-Statement-0001 • 0 points • 28 comments

I've gotten my 3xP40 right down to 6W. Anyone else done any optimizations?

It normally idles at about 116W, but with some hacks/tweaks I managed to get it to suspend automatically and wake on lan and be ready to go in about 20 seconds. I'm running llama3.1 70b.

Comments

u/Worth-Ad-2795 • 1 points • 2024-08-30 21:54 UTC

side question - any software that shows you the W pull without using a physical outlet plug?

u/kryptkpr • 1 points • 2024-08-30 20:29 UTC

<https://preview.redd.it/q13xxibm2vld1.jpeg?width=4080&format=pjpg&auto=webp&s=ec35f52b5aafbfb00e6f120182a2fdf2dab7e788>

It depends what they're doing, but here's a 2x400W at the wall scenario. This is 2xP40 (left) and 2xP100 (right) all running inference. I believe the P40s had a 70B loaded across them while the P100 each had an 8B.

I use custom frontend that can inference multiple completions from multiple models so my power usage can get fairly bananas I've hit 1KW before (there are 2x3060 on that right side that are idle here)

u/Red_Redditor_Reddit • 1 points • 2024-08-30 16:14 UTC

Underclocked 14900k with 4090. 550w for prompting, 330w for text output, 240w if CPU only, 65w idle.

u/segmond • 1 points • 2024-08-30 16:08 UTC

My electric bill is definitely up about \$100+/month. Trying to figure out if it's 6 GPUs or inflation of utility costs.

u/ramzeez88 • 1 points • 2024-08-31 09:40 UTC

I would use some solar power if I was you (if it's possible)

u/ScrapEngineer_ • 1 points • 2024-08-30 17:00 UTC

Def 6 GPU, even more if they are 4090s

u/a_beautiful_rhind • 3 points • 2024-08-30 11:48 UTC

I tried to optimize but it's basically impossible with a server. I pull about 250-300w at idle and it ends up costing \$25-30 a month, just like a subscription.

When picking a board, make sure it has sleep.

u/Some_Endian_FP17 • 2 points • 2024-08-30 11:02 UTC

Do laptop users count?

Snapdragon X1 Surface Pro: maybe 5W idle with a model loaded including the screen and WiFi powered on, 30 to 35W running inference on 6 to 8 cores.

u/No-Statement-0001 • 2 points • 2024-08-30 19:11 UTC

which model are you running and tok/sec. That's impressively low inference power usage.

u/Sambojin1 • 2 points • 2024-08-31 01:20 UTC

About 3.5-5watts on my phone. That's its max power draw anyway. It does chew through a 5000mAh battery, but no more than mobile gaming does.

3-7 tokens a second on ARM optimized models (Gemmasutra 2B to code/ Llama3.1 8B, and everything in between). Not bad for a \$200 phone (a Motorola g84). Better phones would be, umm, better. At probably only 1.5-3x the power draw.

Sure, I'm model size limited, but it's fun from a hobbyist/ vague interest in AI perspective. I can run larger models, but these do stuff quickly enough. I might buy something decent for my next phone, but this does all the normal phone stuff too, so I'm not sure blowing an extra 800\$ just for slightly faster LLMs is worth it.